Year 11 Mathematics

Handling Data 4

4.2 Tally Charts

Tally Charts

STARTER · 5 MIN

Date: _____ **Title:** Tally Charts

LEARNING OBJECTIVES

- Record data in a tally chart
- Complete and use frequency tables
- Group data into class intervals

STARTER *Answer in your book. No calculators.*

1. Work out 5 + 5 + 5 + 2
2. Work out 4 × 5 + 3
3. How many whole numbers from 10 to 19?
4. Is 20 in the group $10 \leq x < 20$?

Starter — Answers

STARTER

Tick or correct in green pen

1. **17**

2. $20 + 3 = \mathbf{23}$

3. **10**

4. **No**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can record data in a tally chart	☐
SILVER	I can complete and use frequency tables	☐
GOLD	I can group data into class intervals	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Tally Charts

INTRODUCTION · 3 MIN

A tally chart records data as you collect it. Count the tallies to get the frequency.

Tally

One line for each item.

Every 5th line crosses the four: a gate

Frequency

How many times each value occurs.

Total = sum of the frequencies

Groups

Use class intervals for lots of values.

0–9, 10–19, 20–29

Inequalities

Continuous data uses ≤ and <.

$10 \le t < 20$ includes 10, not 20

Examples 1 and 2 — Making a Tally Chart

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

Make a tally chart for: red, blue, red, green, red, blue, red, red.

Example 2

Find the total frequency.

Sport	Tally	Frequency
Football	𝍸 |	
Tennis	𝍸 ||||	
Swimming	||||	

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Red 5

Blue 2

Green 1

Colour	Tally	Frequency
Red	𝍸	5
Blue	||	2
Green	|	1

Tip: the 5th tally crosses the first four.

Example 2

6 + 9 + 4

Total **= 19**

Tip: count each row, then add.

Task 1

PRACTICE · 9 MIN

BRONZE *Use the tally chart*

1. How many chose dog?
2. How many chose cat?
3. Which pet was most popular?
4. How many people were asked?
5. How many more chose cat than fish?

★ What fraction chose rabbit?

Pet	Tally
Dog	𝍸 |||
Cat	𝍸 𝍸 |
Fish	||||
Rabbit	𝍸 |
Other	|||

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. **8**

2. **11**

3. **Cat**

4. $8 + 11 + 4 + 6 + 3 = \mathbf{32}$

5. $11 - 4 = \mathbf{7}$

★ $\frac{6}{32} = \frac{\mathbf{3}}{\mathbf{16}}$

Examples 3 and 4 — Frequency Tables

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Example 3

Goals scored in 10 matches: 0, 2, 1, 0, 3, 1, 1, 2, 0, 1. Make a frequency table.

Example 4

Find the total number of goals.

Goals	Frequency
0	3
1	4
2	2
3	1

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Count each value

Goals	Frequency
0	3
1	4
2	2
3	1

Tip: tick off each value as you count it.

Example 4

$0 \times 3 + 1 \times 4 + 2 \times 2 + 3 \times 1$

$0 + 4 + 4 + 3$

Total **= 11**

Tip: multiply each value by its frequency.

Task 2

PRACTICE · 9 MIN

SILVER *Use the frequency table*

1. How many students have 2 siblings?
2. How many students were asked?
3. Which number of siblings is most common?
4. How many have more than 1 sibling?
5. Find the total number of siblings

Siblings	Frequency
0	5
1	9
2	7
3	3
4	1

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. **7**

2. $5 + 9 + 7 + 3 + 1 = \mathbf{25}$

3. **1 sibling**

4. $7 + 3 + 1 = \mathbf{11}$

5. $0 + 9 + 14 + 9 + 4 = \mathbf{36}$

Examples 5 and 6 — Grouped Frequency Tables

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Example 5

(2 marks)

Times (s): 12, 25, 18, 31, 22, 15, 27, 19, 34, 21. Make a grouped frequency table.

Example 6

(2 marks)

How many took at least 20 seconds?

Time t (s)	Frequency
$10 \leq t < 20$	4
$20 \leq t < 30$	4
$30 \leq t < 40$	2

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Put each time in its group

Time t (s)	Frequency
$10 \leq t < 20$	4
$20 \leq t < 30$	4
$30 \leq t < 40$	2

Tip: 20 goes in $20 \leq t < 30$.

Example 6

4 + 2

6 people

Tip: 'at least 20' includes 20.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

1. How many people were measured? (1 mark)

2. Write down the modal class. (1 mark)

3. How many people are shorter than 160 cm? (2 marks)

4. What percentage are at least 170 cm? (1 d.p.) (2 marks)

5. Which class does 150 cm go in? Explain. (1 mark)

Height h (cm)	Frequency
140 ≤ h < 150	3
150 ≤ h < 160	8
160 ≤ h < 170	12
170 ≤ h < 180	5
180 ≤ h < 190	2

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $3 + 8 + 12 + 5 + 2 = \mathbf{30}$

2. $\mathbf{160 \leq h < 170}$

3. $3 + 8 = \mathbf{11}$

4. $7 \div 30 \times 100 = \mathbf{23.3\%}$

5. $\mathbf{150 \leq h < 160}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Two gates and three lines. What is the frequency?

SILVER Frequencies 4, 7, 2, 5. Find the total.

GOLD Which group does 30 go in: $20 \leq x < 30$ or $30 \leq x < 40$?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $5 + 5 + 3 = \mathbf{13}$

SILVER $4 + 7 + 2 + 5 = \mathbf{18}$

GOLD $\mathbf{30 \leq x < 40}$

Exam Practice — Tally Charts

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1

(2 marks)

The tally chart shows favourite fruits.

Fruit	Tally	Frequency
Apple	𝍸 ||	
Banana	𝍸 𝍸 ||	
Orange	||||	
Grape	𝍸 ||||	

(a) Complete the frequency column. (1)

(b) Write down the mode. (1)

.....

Question 2

(2 marks)

Goals scored in 12 matches:

2 0 1 3 1 0 2 1 1 4 0 1

Complete the frequency table.

Goals	0	1	2	3	4
Frequency					

Exam Practice — Frequency Tables

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: answer on the line, show all your working

Question 3

(4 marks)

Siblings of some students:

Siblings	0	1	2	3	4
Frequency	5	9	7	3	1

(a) How many students are there? (1)

.....

(b) How many have 2 or more? (1)

.....

(c) Find the total number of siblings. (2)

.....

Question 4

(3 marks)

Pets of 20 students:

Pets	0	1	2	3
Frequency	6	8	4	2

(a) What fraction have no pets? Give it in its simplest form. (2)

.....

(b) What percentage have at least 2 pets? (1)

..... %

Exam Practice — Grouped Frequency Tables

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: answer on the line, show all your working

Question 5

(3 marks)

Times, t seconds, of 20 runners:

Time (t s)	Frequency
$10 \le t < 20$	4
$20 \le t < 30$	9
$30 \le t < 40$	6
$40 \le t < 50$	1

(a) Write down the modal class. (1)

.....

(b) What % took 30 s or more? (2)

..... %

Question 6

(2 marks)

Masses of 12 parcels (kg):

2.4 5.1 3.8 1.2 4.6 2.9

3.3 5.6 0.8 4.0 2.2 3.5

Complete the frequency table.

Mass (m kg)	Frequency
$0 < m \le 2$	
$2 < m \le 4$	
$4 < m \le 6$	

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 7, 12, 4, 9 — B1

(b) banana — B1

Tip: the mode has the highest frequency.

Question 2

0: 3 1: 5 2: 2 3: 1 4: 1 — B2

B1 for 3 or 4 correct

Tip: check the total is 12.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) 5 + 9 + 7 + 3 + 1 = 25 — B1

(b) 7 + 3 + 1 = 11 — B1

(c) 0 + 9 + 14 + 9 + 4 — M1

= 36 — A1

Tip: total = siblings × frequency, added.

Question 4

(a) $\frac{6}{20} = \frac{3}{10}$ — M1 A1

(b) 6 out of 20 = 30% — B1

Tip: at least 2 means 2 or more.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) 20 ≤ t < 30 — B1

(b) 6 + 1 = 7 out of 20 — M1

= 35% — A1

Tip: '30 s or more' includes 30.

Question 6

2, 7, 3 — B2

B1 for two correct

Tip: ≤ 4 includes 4, so 4.0 is in 2 < m ≤ 4.